

Chronic Kidney Disease (CKD) Prediction — Model Comparison Report

Dataset Summary

Dataset name: CKD.csv
Target column: classification (Yes → CKD, No → Not CKD)
Split: 67% training, 33% testing
Goal: Predict whether a patient has CKD based on multiple clinical and biochemical features.

S.No	Algorithm	Accuracy	Precision	Recall	F1-Score	Remarks
1	Random Forest	0.98	0.98	0.98	0.98	Excellent accuracy and consistency
2	Decision Tree	0.92	0.92	0.93	0.92	Slightly overfits
3	SVM (Linear Kernel)	0.98	0.98	0.98	0.98	Performs equally well
4	SVM (RBF Kernel)	0.62	0.38	0.62	0.47	Poor performance
5	Logistic Regression	0.90	0.91	0.90	0.90	Good baseline model
6	K-Nearest Neighbors	0.94	0.95	0.94	0.94	Strong, but slower
7	Naive Bayes	0.83	0.87	0.83	0.83	Less effective for this dataset

Final Model Selection — Random Forest Classifier

Random Forest Classifier was selected as the best performing model with an accuracy of **98%**. It achieved balanced Precision, Recall, and F1-Score, and handled mixed data types effectively.

Justification:

- Highest accuracy and consistency across metrics.
- Robust to noise and overfitting.
- Provides interpretability through feature importance.
- Performs well with categorical and numerical medical data.

Confusion Matrix (Random Forest)

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| Actual / Predicted | CKD (Yes) | Not CKD (No) |
|-----|-----|-----|
| CKD (Yes) | 81 | 1 |
| Not CKD (No) | 1 | 50 |
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Conclusion:

After testing multiple classification algorithms, the Random Forest Classifier demonstrated the best trade-off between accuracy, generalization, and interpretability. With 98% accuracy and balanced precision/recall, it is the recommended model for CKD prediction and clinical decision support.